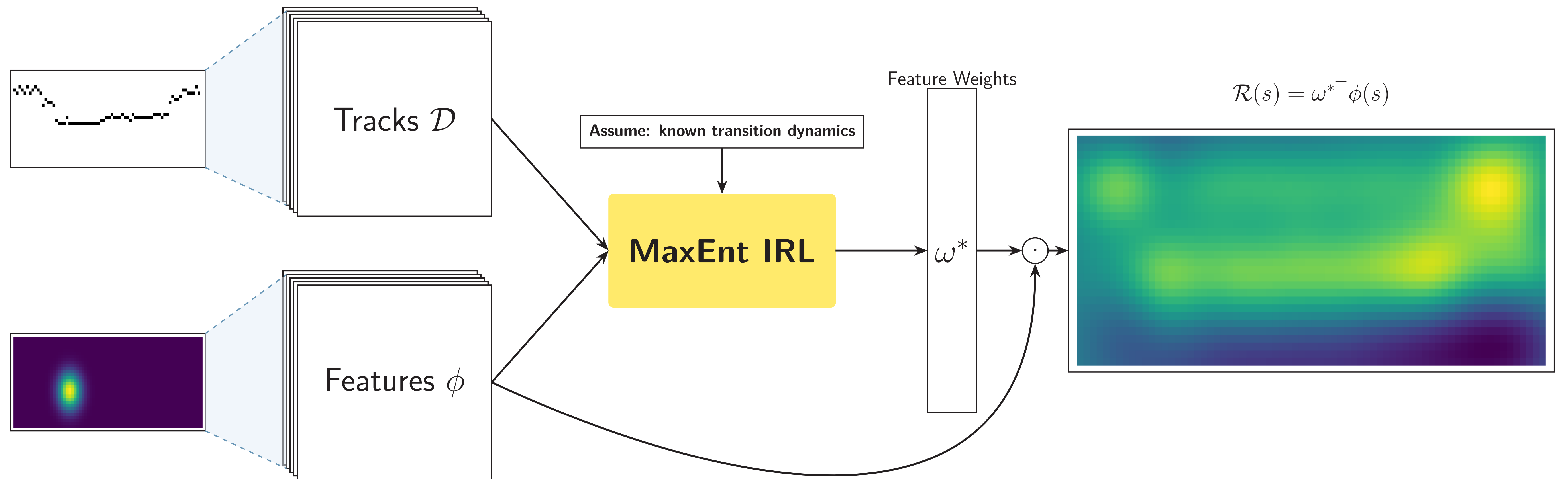


A brief (more) mathematical explanation of Maximum Entropy IRL

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Maximum Entropy Inverse Reinforcement Learning



Graphical Intuition for the application of Maximum Entropy Inverse Reinforcement Learning: The algorithm uses the dataset of trajectories $\mathcal{D}$, the features ϕ and the assumed transition dynamics of the Markov Decision Process to calculate the optimal feature weights ω^* , which can in turn be multiplied with the features to obtain a reward map $\mathcal{R}$.

Maximum Entropy Inverse Reinforcement Learning (MaxEnt) is an algorithm for solving an inverse reinforcement learning problem, first presented by Ziebart et al. in 2008⁽¹⁾. It has two core ideas, **feature expectation matching** and **entropy maximization**, both of which will be explained in the following:

Goal. The goal of MaxEnt is to recover a reward $\mathcal{R}$ from a set of demonstrations $\mathcal{D} = \{\tau_i\}_{i=1}^n$, consisting of trajectories $\tau = ((s_1, a_1), (s_2, a_2), \dots, (s_{n-1}, a_{n-1}), s_n)$, in a Markov Decision Process (MDP) with known dynamics. Knowing the dynamics means that, for taking action a in state s_i , the transition probabilities to next states s_{i+1} are known. Here, we assume deterministic transitions, i.e. the action "move up two metres" always takes fish up two metres in the next step, without any element of stochasticity in their movement. Assume $\mathcal{R}(\tau) = \omega^\top \phi(\tau)$ with $\phi(\tau) = \sum_t \phi(s_t)$. The first key idea of MaxEnt is that the **learner's expected feature utilization must match the expert's**:

$$\begin{aligned}\mathbb{E}_{\pi^L}[\phi(\tau)] &= \mathbb{E}_{\pi^E}[\phi(\tau)] \\ \mathbb{E}_{\pi^L}[\phi(\tau)] &= \sum_{\tau} p_{\pi^L}(\tau) \phi(\tau)\end{aligned}$$

In practice, we use our best guess of the expert's feature expectation - the average feature counts $\frac{1}{N} \sum_i \phi(\tau_i)$. Because many distributions of the probabilities of trajectories occurring (the $p(\tau)$) satisfy this, we choose the one with **maximum entropy**, i.e. the one that is least committed to the data. While unintuitive at first, this is similar to how we avoid overfitting in other statistical contexts: a good explanation leaves the most possible room for unseen or new data, while fulfilling the criteria set out. Mathematically, maximizing the entropy $\mathcal{H}$ then yields a constrained optimization problem:

$$\arg \max_p \mathcal{H}(p) \quad \text{subject to} \quad \mathbb{E}_{\pi^L}[\phi(\tau)] = \mathbb{E}_{\pi^E}[\phi(\tau)]; \quad \sum_{\tau} p(\tau) = 1; \quad \forall \tau : p(\tau) > 0$$

Solution. Solving this problem (Lagrange multipliers) gives

$$p(\tau | \omega) = \frac{1}{Z(\omega)} \exp(\omega^\top \phi(\tau)); \quad Z(\omega) = \sum_{\tau} \exp(\omega^\top \phi(\tau)),$$

so higher-reward trajectories are exponentially more likely, and ω are the Lagrange multipliers.

Learning. The Lagrangian dual of this problem is exactly maximum likelihood of the demonstrations under $p(\tau | \omega)$:

$$\begin{aligned}\omega^* &= \arg \max_{\omega} \mathcal{L}(\omega), \quad \mathcal{L}(\omega) = \frac{1}{N} \sum_{i=1}^N \log p(\tau_i | \omega), \\ \nabla_{\omega} \mathcal{L} &= \underbrace{\frac{1}{N} \sum_i \phi(\tau_i)}_{\text{expert (data)}} - \underbrace{\mathbb{E}_{p(\tau|\omega)}[\phi(\tau)]}_{\text{learner}}.\end{aligned}$$

$\mathcal{L}$ is concave, and the gradient vanishes exactly when the features match. The learner's expectation is computed with known dynamics via soft value iteration and state visitation frequencies, $\mathbb{E}_p[\phi(\tau)] = \sum_s D_{\omega}(s) \phi(s)$.

Result. The result of the algorithm after convergence is the weights-vector ω^* . It is "chosen" such that the trajectories in the dataset are maximally likely, and multiplied with the features ϕ we can obtain a reward map $\mathcal{R}$.

References & Further Reading

References

(1) Ziebart, B. D.; Maas, A.; Bagnell, J. A.; Dey, A. K. Maximum Entropy Inverse Reinforcement Learning. 2008.

Further Reading

A super informative write-up by Maximilian Luz going into yet more detail: <https://mxnluz.io/blog/posts/2022/08/irl-maxent>